

Ian Kaufman

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EDUCATION

Stony Brook University

Expected Graduation: May 2027

Bachelor of Science in Computer Science & Applied Math and Statistics

GPA: 3.64 / 4.00

Relevant Coursework: Software Development, Data Structures, Analysis of Algorithms, Scripting Languages, Object-Oriented Programming, Data Science, Probability & Statistics, Artificial Intelligence, Operating Systems, Computational Theory, Calculus 3

Teaching Assistant: Finite Mathematical Structures; Technology in Media

SKILLS

Languages: Java, Python, C, SQL, R, JavaScript, HTML/CSS, Linux/UNIX Shell Scripting

Frameworks & Libraries: Django, FastAPI, Flask, SQLAlchemy, React, Node.js/Express, Pytest, NumPy, Pandas, Matplotlib

Tools & Platforms: Git/GitHub, Docker, Docker Compose, PostgreSQL, MongoDB, Uvicorn, VS Code, GitHub Actions (CI/CD)

Cloud & Infrastructure: AWS Technical Essentials Certification; EC2, S3, Lightsail, Azure App Service, GCP (BigQuery, Firestore)

PROFESSIONAL EXPERIENCE

Amazon Web Services

Portland, OR

Incoming Software Development Engineer Intern

Fall 2026

- Incoming SDE Intern on AWS, selected for a Fall 2026 internship after completing Amazon's technical interview process.

Ally Financial

Charlotte, NC

Technology Intern—Systems Architecture (ETO)

May 2026 - Present

- Saved an estimated **40+ hours annually**, reduced manual effort by **30%**, and improved firewall request workflow speed by **25%** by developing an approved **Splunk and ServiceNow API** automation service for IP/subnet traffic lookup, flow selection, and formatted access request generation.
- Preserved **500 limited Nutanix project categories** and improved metadata visibility by an estimated **60%** by designing a **metadata-to-JSON conversion utility** that injects structured JSON directly into projects, replacing category-based tagging with a more scalable metadata approach.

MyStage Music

Remote

Software Engineering Intern

Jan 2026–May 2026

- Architected and deployed an end-to-end deduplication and automerger pipeline processing **300K+ Firestore records**, using Firebase, Pydantic, BigQuery SQL transformations, and Firestore writeback to surface validated results in the admin UI.
- Containerized the deduplication and automerger pipeline with Docker for **Google Cloud deployment**, enabling weekly runs to refresh validated Firestore records automatically.

Springer Capital

Remote

Backend Software Engineering Intern

Jul 2025–Oct 2025

- Designed a normalized PostgreSQL schema (employees, departments, roles) with foreign keys, unique constraints, and composite/partial indexes aligned to high-frequency queries; shipped migrations via Django.
- Refactored backend data-access patterns to reduce inefficient database queries, improve API response consistency by decreasing **p95 latency 30%**, and support cleaner reporting over employee and department records.

PROJECTS

Book Recommender | <https://iansbookrecs.com> | Python, FastAPI, JavaScript, scikit-learn, AWS

- Deployed a full-stack book recommendation app on AWS Lightsail using FastAPI, SQLite, Docker, and a vanilla JavaScript frontend, serving **20+ users** with account-based libraries and personalized recommendations.
- Engineered a **content-based recommendation engine** blending **IDF-weighted title/description similarity** with metadata enrichment from Open Library, language, author/book and genre filtering, and series-volume collapsing.
- Built **per-user accounts** with **PBKDF2-hashed passwords** and session-cookie authentication, a personal library with user-defined sections and reading statuses driving **scoped recommendations**, and a **thread-safe LRU cache** that skips unchanged recommendation runs.

Distributed Task Queue | github.com/Iank314/task-queue | aiohttp • PostgreSQL • Python • Docker Compose • GitHub Actions

- Designed a distributed task queue with an **aiohttp broker**, **PostgreSQL-backed job store**, horizontally scalable worker pool, and row-level locking, preventing duplicate job execution across concurrent workers through atomic job claiming.
- Improved queue reliability and observability across **4+ operational metrics** by implementing priority scheduling, exponential backoff retries, a dead-letter queue, and tracking for queue depth, throughput, job states, and failure rates.

LEADERSHIP

Vice President & Co-Founder

Stony Brook University

Business Analytics Club

Jan 2026 – Present

- Co-founded and led** a student organization with 30+ members focused on analytics and data-driven decision making.
- Organized industry speaker events with professionals from finance/tech firms such as **Alvarez & Marsal**; 50+ in attendance.